

CMS Surgery Form 1 - Answers & Explanations

1. A) Appropriate in view of the medical necessity to remove the inflamed appendix
 - Acutely inflamed appendix is a medical emergency and therefore doesn't require consent.
2. A) Aldosteronoma
 - Hyperaldosteronism → hypokalemia
 - Aldosterone escape → normal Na levels
3. C) Ruptured spleen
 - Kehr sign: referred pain in shoulder due to irritants (eg, blood) in the peritoneal cavity
 - Kehr sign in the left shoulder is a classic sxs of a ruptured spleen
4. B) High-output congesting heart failure
 - Circulatory overload → ↑ diastolic pressure in LV → pulmonary edema
5. D) Squamous cell carcinoma
 - SCC → PTHrP → hypercalcemia
6. D) Needle-localized open biopsy
 - Percutaneous core biopsy, localization with needle and surgically sampling, is preferred for evaluation of microcalcifications
 - Fine-needle aspiration is inadequate as it may miss the cancer cells
7. B) Cardiogenic shock
 - SOB, JVD, hypotension, hepatomegaly, ST elevation in anterior leads
8. E) Testicular tumor
 - Young age, painless progressive enlargement over 8 months = cancer
 - Hydrocele and Spermatocoele both transilluminate
 - Epididymitis is painful
9. C) Rapid infusion of 0.9% saline
 - Patient is unstable
10. C) Cytomegalovirus
 - Colitis in HIV + nuclear inclusion bodies = CMV
 - Presents with hematochezia, abdominal pain, weight loss, low-grade fever
11. E) Lactated Ringer solution
 - Ranson's criteria → pancreatitis
 - Rx: fluids, analgesics, NPO
12. G) Sphenopalatine artery
 - Sphenopalatine artery is the MCC of posterior nosebleeds
 - 90% of nosebleeds are anteriorly located at Kiesselbach's plexus
13. D) 15 L of lactated Ringer solution
 - Parkland formula: $4 \times \text{weight in kg} \times \% \text{ body surface area that is burnt}$
 - $4 \times 75 \times 50 = 15000 \text{ ml} = 15\text{L}$
14. B) Bladder outflow obstruction
 - Old male with mass in lower quadrant indicated BPH
15. C) Fat embolism
 - Associated with long bone fractures and liposuction
 - Triad: Neurologic dysfunction, petechial rash (usually in axilla), respiratory distress (hypoxemia)
 - Therapy is supportive

16. D) *Pseudomonas* species
- Gram negative rod
17. E) Incision and drainage of the lesion
- Abscesses must be drained
18. E) Surgical decompression
- Compartment syndrome
 - Paresthesia and pain on passive stretch are early sxs
19. D) Lack of normal joint sensation
- Neuropathic (Charcot) arthropathy
20. A) Aortic arteriography
- Mild tenderness over the chest and a 'somewhat' widened mediastinum warrant an aortic arteriography
 - If the patient had presented with "tearing chest pain radiating to the back" and a widened mediastinum following a MVC, then an immediate thoracotomy would be indicated
21. E) Water-soluble contrast upper gastrointestinal study
- Rupture/perforation is a complication of esophagoscopy
 - Water-soluble (gastrograffin) contrast study can diagnose rupture
22. E) Initiation of dialysis
- 1 urea nitrogen must be corrected pre-operatively because 1 urea → platelet dysfunction → hemorrhage during surgery
 - Decompression (DDAVP) → ↑ release of factor VIII/vWf multimers, which can correct the coagulopathy in uremic patients
23. B) Low-dose heparin prophylaxis **(not sure)**
- Heparin is best choice for pre-surgical VTE prophylaxis because it has a short half-life (~45 mins) and is easily reversed with protamine sulfate
24. A) Epidural hematoma
- The middle meningeal artery runs underneath the pterion, the weak region of the skull where the frontal, parietal, temporal, and sphenoid bones join together
 - Traumatic blows to this area classically cause an epidural hematoma
25. B) Femoral hernia
- Protrudes through femoral canal lateral to the pubic tubercle and medial to the femoral vessels
26. G) Sliding hiatal hernia
- Sliding hiatal hernias account for >90% of hiatal hernias
 - Associated with GERD and reflux esophagitis → ↑ risk of Barrett's esophagus
27. C) Failure to suppress class II antigen recognition in the host
- Acute rejection has an onset of weeks to months
 - CD4 against MHC class II antigen
28. E) Increased serum aldosterone concentration
- Renin increases on the affected side due to ↓ blood supply → ↑ aldosterone
 - Blood supply to unaffected side is normal, so no need to release renin
29. B) Glucagonoma **(not sure)**
- Glucagonomas: diarrhea, dermatitis, declining weight which are some of the (D)s of glucagonoma.
 - Insulinoma presents with recurrent fasting hypoglycemia
 - VIPoma is associated with MEN 1 syndrome and presents with WDHA syndrome (Watery Diarrhea, Hypokalemia, Achlorhydria)

30. H) Increase positive end-expiratory pressure
- Normal PO₂ is between 80-100
 - FiO₂ and PEEP determine PO₂;
 - Tidal volume and RR determine PCO₂
31. D) Proctocolectomy with an ileoanal reservoir
- Prophylactic colectomy is the best long-term management in FAP because 100% of cases will eventually progress to CRC
 - FAP patients should be screened with a colonoscopy every year starting at age 10-12
32. B) Clostridium perfringens
- Crepitus - gas-forming = clostridium
33. E) Laparotomy
- Presence of peritoneal signs in a patient with acute abdomen = indication for ex-lap
 - Presence of involuntary guarding and rigidity is a contraindication for colonoscopy or sigmoidoscopy
34. I) Thoracic spinal cord
- Breast cancer metastasis to spinal cord
 - Decreased sensation below T4 indicates a thoracic location
35. D) Heparin-induced thrombocytopenia
- Antibody-heparin-PF4 complex activates platelets → thrombosis and thrombocytopenia
36. E) RBC 30/hpf (in sediment)
- Hematuria, gross or microscopic, occurs in the majority of patients with symptomatic nephrolithiasis
37. C) Atrial myxoma
- Obstruction of mitral valve in diastole → diastolic murmur
 - Changing of positions → movement of tumor → disappearance of murmur
 - Cardiac myxoma produce IL-6 → fever and weight loss
38. E) Gastric bypass
- This patient has comorbidities due to morbid obesity (she is most likely bed-ridden → poor personal hygiene)
 - BMI > 35 with a major comorbidity (eg, HTN, DM, etc.) qualifies for bariatric surgery; BMI > 40 qualifies without the presence of comorbidities
39. F) Ovarian adenocarcinoma
- Nulligravid → uninterrupted menstrual cycles (follicular rupture and tissue damage) → ↑ risk of ovarian cancer
 - Advanced ovarian cancer → peritoneal seeding and inflammation → ascites
 - Meigs syndrome: triad of ovarian fibroma, ascites, and pleural effusion
40. C) Lymphoma
- Lymphatic obstruction (eg, neoplasm) → chylous ascites
41. D) Formation of antiplatelet antibodies
- Immune thrombocytopenia (ITP) causes destruction of platelets in the spleen due to anti-GpIIb/IIIa antibodies
 - Labs: ↑ megakaryocytes, ↓ platelets
 - Treatment: steroids, IVIG, rituximab, splenectomy for refractory cases
42. A) CT scan of the abdomen with contrast
- Location of injury and gross hematuria suggest kidney injury
 - Contrast is really excreted and is used to visualize the collecting system

43. D) Intravenous administration of calcium gluconate
- Hypocalcemia after parathyroidectomy is transient, but IV calcium should be given to protect the heart
44. D) Slipped capital femoral epiphysis
- Classically presents in teens/pre-teens that are obese or above 90th percentile in weight
45. B) Carotid duplex ultrasonography
- Bilateral carotid bruits indicate stenosis in the carotids → compromised blood flow → cerebral ischemia → TIA
 - Carotid U/S should be performed to determine the level of stenosis and the corresponding treatment
46. A) Avascular necrosis of the femoral head
- Legg-Calvé-Perthes disease
 - Common in males between 5-8 years old
47. E) Malignant pleural effusion
- Past hx of cancer, respiratory distress, cyanotic, cachectic, decreased breath sounds, dullness to percussion
 - Pleural friction rub could be due to post-radiation fibrosis of the lungs
48. D) Cholangitis, ascending
- Charcot's triad: jaundice, fever, RUQ pain
 - Reynolds pentad: Charcot's triad, altered mental status, hypotension
 - ↑ alkaline phosphatase and ↑ conjugated bilirubin
49. A) Determination of ankle brachial indices
- ABI is the preferred initial step in diagnosis of peripheral arterial disease (PAD)
 - Arterial U/S of the lower extremities is less sensitive and specific than ABI for the diagnosis
50. G) Melanoma
- Subungual hematoma, unlike melanoma, will gradually clear as the nail heals
 - Subungual hematoma will form a transverse groove across the surface as the nail heals